General Equation of Second Degree

1. **Discuss the condition of general equation of the second degree i. a pair of straight lines, ii. Two parallel lines, iii. a circle, iv. a parabola, v. an ellipse and vi. a hyperbola.**

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ will represent

(1) a pair of straight lines if the determinant $\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

(i) two parallel lines if $\Delta = 0,\ ab = h^2$

(ii) two perpendicular lines if $\Delta = 0,\ a + b = 0$

(2) a circle if $a = b,\ h = 0$

(3) a parabola if $\Delta \neq 0,\ ab = h^2$

(4) an ellipse if $\Delta \neq 0,\ ab - h^2 > 0$

(5) a hyperbola if $\Delta \neq 0,\ ab - h^2 < 0$

(6) a rectangular hyperbola if $\Delta \neq 0, a\,b - h^2 < 0,\ a + b = 0$

2 The standard equation of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

The standard equation of an hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

The standard equation of an parabola $y^2 = 4ax$

3 **Test the nature of the conic given by the equation $12x^2 - 10xy + 2y^2 + 11x - 5y + 2 = 0$.**

Solution:

General equation of second degree is

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$

Given equation is

$12x^2 - 10xy + 2y^2 + 11x - 5y + 2 = 0$ ……. (ii)

Comparing equation (i) and (ii), we get

$a = 12, h = -5, b = 2, g = \frac{11}{2}, f = -\frac{5}{2}, c = 2$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$\Rightarrow \Delta = 12.2.2 + 2.-\frac{5}{2}.\frac{11}{2}.(-5) - 12.\left(-\frac{5}{2}\right)^2 - 2.\left(\frac{11}{2}\right)^2 - 2(-5)^2$

$\Rightarrow \Delta = 48 + \frac{25.11}{2} - 12.\frac{25}{4} - 2.\frac{121}{4} - 2.25$

$\Rightarrow \Delta = 48 + \frac{275}{2} - 75 - \frac{121}{2} - 50$

$\Rightarrow \Delta = -77 + \frac{275}{2} - \frac{121}{2}$

$\Rightarrow \Delta = \frac{-154 + 275 - 121}{2}$

$\Rightarrow \Delta = \frac{-275 + 275}{2}$

$\Rightarrow \Delta = 0$

So this given equation can be represents as a pair of straight lines.

4 **Test the nature of the conic given by the equation $4x^2 - 4xy + y^2 - 12x + 6y + 9 = 0$.**

Solution:

General equation of second degree is

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ ……. (i)

Given equation is

$4x^2 - 4xy + y^2 - 12x + 6y + 9 = 0$. ……. (ii)

Comparing equation (i) and (ii), we get

$a = 4, h = -2, b = 1, g = -6, f = 3, c = 9$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$\Rightarrow \Delta = 4.1.9 + 2.3.(-6).(-2) - 4.(3)^2 - 1.(-6)^2 - 9(-2)^2$

$\Rightarrow \Delta = 36 + 72 - 36 - 36 - 36$

$\Rightarrow \Delta = 0$

So this given equation can be represents as a pair of straight lines.

$ab - h^2 = 4.1 - (2)^2 = 4 - 4 = 0$

$\because \ \Delta = 0 \, , ab - h^2 = 0$

So the given equation represents two parallel lines.

5 **Test the nature of the conic given by the equation $6x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0$.**

Solution:

General equation of second degree is

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$

Given equation is

$6x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0. \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (ii)$

Comparing equation (i) and (ii), we get

$a = 6, h = -\frac{5}{2}, b = -6, g = 7, f = \frac{5}{2}, c = 4$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$\Rightarrow \Delta = 6.(-6).4 + 2.\frac{5}{2}.7.\left(-\frac{5}{2}\right) - 6.\left(\frac{5}{2}\right)^2 + 6.(7)^2 - 4\left(-\frac{5}{2}\right)^2$

$\Rightarrow \Delta = -144 - \frac{175}{2} - 6.\frac{25}{4} - 6.49 - 4.\frac{25}{4}$

$$\Rightarrow \Delta = -144 - \frac{175}{2} - \frac{75}{2} - 294 - 25$$

$$\Rightarrow \Delta = 125 - \frac{175}{2} - \frac{75}{2}$$

$$\Rightarrow \Delta = \frac{250 - 175 - 75}{2}$$

$$\Rightarrow \Delta = \frac{270 - 270}{2}$$

$$\Rightarrow \Delta = 0$$

So this given equation can be represents as a pair of straight lines.

$$ab - h^2 = 6.(-6) - \left(-\frac{5}{2}\right)^2 = -36 - \frac{25}{4} = \frac{-144 - 25}{2} \neq 0$$

$$a + b = 6 + (-6) = 0$$

$\because\ \Delta = 0\ , a + b = 0$

So the given equation represents two perpendicular lines.

6. Let the equation

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$$

Reduce process

Put $x = x' + \alpha$ and $y = y' + \beta$

Putting the values of x and y in (i), we get

$$a(x' + \alpha)^2 + 2h(x' + \alpha)(y' + \beta) + b(y' + \beta)^2 + 2g(x' + \alpha) + 2f(y' + \beta) + c = 0$$

$$\Rightarrow a(x'^2 + 2\alpha x' + \alpha^2) + 2h(x'y' + x'\beta + \alpha x' + \alpha\beta) + b(y'^2 + 2\beta y' + \beta^2) + 2gx' + 2g\alpha + 2fy' + 2f\beta + c = 0$$

$$\Rightarrow ax'^2 + 2a\alpha x' + a\alpha^2 + 2hx'y' + 2hx'\beta + 2h\alpha x' + 2h\alpha\beta + by'^2 + 2b\beta y' + b\beta^2 + 2gx' + 2g\alpha + 2fy' + 2f\beta + c = 0$$

$$\Rightarrow ax'^2 + 2hx'y' + by'^2 + 2(a\alpha + h\beta + g)x' + 2(h\alpha + b\beta + f)y' + a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (ii)$$

Then $a\alpha + h\beta + g = 0$

$h\alpha + b\beta + f = 0$

and $a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c = 0$ becomes

$\Rightarrow \alpha(a\alpha + h\beta + g) + \beta(h\alpha + b\beta + f) + g\alpha + f\beta + c = 0$

$\Rightarrow g\alpha + f\beta + c = 0$

Where $c_1 = g\alpha + f\beta + c$

Then equation (ii) becomes we get

$ax'^2 + 2hx'y' + by'^2 + c_1 = 0$

7. **Reduce the equation $32x^2 + 52xy - 7y^2 - 64x - 52y - 148 = 0$ to the standard form and identify it.**

Solution:

General equation of second degree is

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$

Given equation is

$32x^2 + 52xy - 7y^2 - 64x - 52y - 148 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (ii)$

Comparing equation (i) and (ii), we get

$a = 32, h = 26, b = -7, g = -32, f = -26, c = -148$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$\Delta = 32.(-7).(-148) + 2.(-26).(-32).26 - 32.(-26)^2 + 7.(-32)^2 + 148(26)^2$

$\Delta = 33152 + 43264 - 21632 + 7168 + 100048$

$\Delta = 460400$

$\Delta \neq 0$

Now

$ab - h^2 = 32.(-7) - (26)^2$

$$= -224 - 676$$

$$= -900$$

$ab - h^2 < 0$

So this given equation can be represents as a hyperbola.

Let $F(x, y) = 32x^2 + 52xy - 7y^2 - 64x - 52y - 148 = 0$

$$\frac{\delta F}{\delta x} = 64x + 52y - 64 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (iii)$$

$$\frac{\delta F}{\delta y} = 52x - 14y - 52 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (iv)$$

Solving equation (iii) and (iv), we get

$\therefore x = 1, \quad y = 0$

Centre is at $(1, 0)$

New constant, $c_1 = gx + fy + c$

$$= -32.1 - 26.0 - 148$$

$$= -32 - 148$$

$$= -180$$

Therefore, the equation is

$32x^2 + 52xy - 7y^2 - 180 = 0$

When the xy term is removed by rotation of axes. Let the reduced equation be

$a'x^2 + b'y^2 = 180 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (v)$

We know, $a' + b' = a + b = 32 - 7 = 25$

$$\therefore a' + b' = 25$$

$a'b' = ab - h^2 = 32.(-7) - (26)^2 = -900$

$(a' - b')^2 = (a' + b')^2 - 4a'b'$

$= 625 + 3600$

$= 4225$

$\therefore a' - b' = 65$

Now solving $a' + b' = 25$ and $a' - b' = 65$ we get

$a' = 45, b' = -20$

Putting the values a' and b' in (v), we get

$45x^2 - 20y^2 = 180$

$\Rightarrow \frac{x^2}{4} - \frac{y^2}{9} = 1$

$\Rightarrow \frac{x^2}{(2)^2} - \frac{y^2}{(3)^2} = 1$

The given equation represents a hyperbola.

8. **Reduce the equation $x^2 + 12xy - 4y^2 - 6x + 4y - 9 = 0$ to the standard form and identify it.**

Solution:

General equation of second degree is

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$

Given equation is

$x^2 + 12xy - 4y^2 - 6x + 4y - 9 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (ii)$

Comparing equation (i) and (ii), we get

$a = 1, h = 6, b = -4, g = -3, f = 2, c = -9$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$\Delta = 1.(-4).(-9) + 2.2.(-3).6 - 1.(2)^2 + 4.(-3)^2 + 9.(6)^2$

$\Delta = 36 - 72 - 4 + 36 + 324$

$\Delta = 320$

$\Delta \neq 0$

Now

$ab - h^2 = 1.(-4) - (6)^2$

$= -4 - 36$

$= -40$

$ab - h^2 < 0$

So this given equation can be represents a hyperbola.

Let $F(x, y) = x^2 + 12xy - 4y^2 - 6x + 4y - 9 = 0$

$$\frac{\delta F}{\delta x} = 2x + 12y - 6 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (iii)$$

$$\frac{\delta F}{\delta y} = 12x - 8y + 4 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (iv)$$

Solving equation (iii) and (iv), we get

$\therefore x = 0, \quad y = \frac{1}{2}$

Centre is at $\left(0, \frac{1}{2}\right)$

New constant, $c_1 = gx + fy + c$

$= -3.0 + 2.\frac{1}{2} - 9$

$= 1 - 9$

$= -8$

Therefore, the equation is

$x^2 + 12xy - 4y^2 - 8 = 0$

When the xy term is removed by rotation of axes. Let the reduced equation be

$a'x^2 + b'y^2 = 8 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (v)$

We know, $a' + b' = a + b = 1 - 4 = -3$

$\therefore a' + b' = -3$

$a'b' = ab - h^2 = 1.(-4) - (6)^2 = -40$

$(a' - b')^2 = (a' + b')^2 - 4a'b'$

$= 9 + 160$

$= 169$

$\therefore a' - b' = 13$

Now solving $a' + b' = -3$ and $a' - b' = 13$ we get

$a' = 5, b' = -8$

Putting the values a' and b' in (v), we get

$5x^2 - 8y^2 = 8$

$$\Rightarrow \frac{x^2}{\frac{8}{5}} - \frac{y^2}{1} = 1$$

$$\Rightarrow \frac{x^2}{\left(\sqrt{\frac{8}{5}}\right)^2} - \frac{y^2}{(1)^2} = 1$$

The given equation represents a hyperbola.

9. **Reduce the equation $x^2 - 5xy + y^2 + 8x - 20y + 15 = 0$ to the standard form and identify it.**

Solution:

General equation of second degree is

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$

Given equation is

$x^2 - 5xy + y^2 + 8x - 20y + 15 = 0 \ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (ii)$

Comparing equation (i) and (ii), we get

$a = 1, h = \frac{-5}{2}, b = 1, g = 4, f = -10, c = 15$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$\Delta = 1.1.15 + 2(-10).4.\left(\frac{-5}{2}\right) - 1.(-10)^2 - 1.(4)^2 - 15.\left(\frac{-5}{2}\right)^2$

$\Delta = 15 + 200 - 100 - 16 - \frac{375}{4}$

$\Delta = 99 - \frac{375}{4}$

$\Delta = \frac{21}{4}$

$\Delta \neq 0$

Now

$ab - h^2 = 1.1 - \left(\frac{-5}{2}\right)^2$

$= 1 - \frac{25}{4}$

$= -\frac{21}{4}$

$ab - h^2 < 0$

So this given equation can be represents a hyperbola.

Let $F(x, y) = x^2 - 5xy + y^2 + 8x - 20y + 15 = 0$

$$\frac{\delta F}{\delta x} = 2x - 5y + 8 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (iii)$$

$$\frac{\delta F}{\delta y} = -5x + 2y - 20 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (iv)$$

Solving equation (iii) and (iv), we get

$\therefore x = -4, \quad y = 0$

Centre is at $(-4, 0)$

New constant, $c_1 = gx + fy + c$

$$= 4.(-4) + (-10).0 + 15$$

$$= -16 + 15$$

$$= -1$$

Therefore, the equation is

$x^2 - 5xy + y^2 - 1 = 0$

When the xy term is removed by rotation of axes. Let the reduced equation be

$a'x^2 + b'y^2 = 1 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (v)$

We know, $a' + b' = a + b = 1 + 1 = 2$

$$\therefore a' + b' = 2$$

$$a'b' = ab - h^2 = 1.1 - \left(\frac{-5}{2}\right)^2 == -\frac{21}{4}$$

$(a' - b')^2 = (a' + b')^2 - 4a'b'$

$$= 4 - 4.\left(-\frac{21}{4}\right)$$

$$= 25$$

$\therefore a' - b' = 5$

Now solving $a' + b' = 4$ and $a' - b' = 5$ we get

$a' = \frac{7}{2}, b' = -\frac{3}{2}$

Putting the values a' and b' in (v), we get

$\frac{7}{2}x^2 - \frac{3}{2}y^2 = 1$

$\Rightarrow \frac{x^2}{\frac{2}{7}} - \frac{y^2}{\frac{2}{3}} = 1$

$\Rightarrow \frac{x^2}{\left(\sqrt{\frac{2}{7}}\right)^2} - \frac{y^2}{\left(\sqrt{\frac{2}{3}}\right)^2} = 1$

The given equation represents a hyperbola.

10. **What conic does the equation $5x^2 - 2xy + 5y^2 - 8x - 8y - 8 = 0$ represent? Reduce the equation to standard form.**
Solution:

General equation of second degree is

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$

Given equation is

$5x^2 - 2xy + 5y^2 - 8x - 8y - 8 = 0 \ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (ii)$

Comparing equation (i) and (ii), we get

$a = 5, h = -1, b = 5, g = -4, f = -4, c = -8$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$\Delta = 5.5(-8) + 2(-4).(-4).(-1) - 5.(-4)^2 - 5.(-4)^2 + 8.(-1)^2$

$\Delta = -200 - 32 - 80 - 80 + 8$

$\Delta = -384$

$\Delta \neq 0$

Now

$ab - h^2 = 5.5 - (-1)^2$

$= 25 - 1$

$= 24$

$ab - h^2 > 0$

So this given equation can be represents an ellipse.

Let $F(x, y) = 5x^2 - 2xy + 5y^2 - 8x - 8y - 8 = 0$

$$\frac{\delta F}{\delta x} = 10x - 2y - 8 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (iii)$$

$$\frac{\delta F}{\delta y} = -2x + 10y - 8 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (iv)$$

Solving equation (iii) and (iv), we get

$\therefore x = 1, \quad y = 1$

Centre is at $(1, 1)$

New constant, $c_1 = gx + fy + c$

$= (-4).1 + (-4).1 - 8$

$= -4 - 4 - 8$

$= -16$

Therefore, the equation is

$5x^2 - 2xy + 5y^2 - 16 = 0$

When the xy term is removed by rotation of axes. Let the reduced equation be

$a'x^2 + b'y^2 = 16 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (v)$

We know, $a' + b' = a + b = 5 + 5 = 10$

$\therefore a' + b' = 10$

$a'b' = ab - h^2 = 5.5 - (-1)^2 = 24$

$(a' - b')^2 = (a' + b')^2 - 4a'b'$

$= 100 - 4.24$

$= 4$

$\therefore a' - b' = 2$

Now solving $a' + b' = 10$ and $a' - b' = 2$ we get

$a' = 6, b' = 4$

Putting the values a' and b' in (v), we get

$6x^2 + 4y^2 = 16$

$$\Rightarrow \frac{x^2}{\frac{8}{3}} + \frac{y^2}{4} = 1$$

$$\Rightarrow \frac{x^2}{\left(\sqrt{\frac{8}{3}}\right)^2} + \frac{y^2}{(2)^2} = 1$$

The given equation represents an ellipse.

11. **What conic does the equation $8x^2 + 4xy + 5y^2 - 24x - 24y = 0$ represent? Reduce the equation to standard form.**
Solution:

General equation of second degree is

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$

Given equation is

$8x^2 + 4xy + 5y^2 - 24x - 24y = 0 \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (ii)$

Comparing equation (i) and (ii), we get

$a = 8, h = 2, b = 5, g = -12, f = -12, c = 0$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$\Delta = 8.5.0 + 2(-12).(-12).2 - 8.(-12)^2 - 5.(-12)^2 + 0.(2)^2$

$\Delta = 576 - 1240 - 720$

$\Delta = -1394$

$\Delta \neq 0$

Now

$ab - h^2 = 8.5 - (2)^2$

$= 40 - 4$

$= 36$

$ab - h^2 > 0$

So this given equation can be represents an ellipse.

Let $F(x, y) = 8x^2 + 4xy + 5y^2 - 24x - 24y = 0$

$$\frac{\delta F}{\delta x} = 16x + 4y - 24 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (iii)$$

$$\frac{\delta F}{\delta y} = 4x + 10y - 24 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (iv)$$

Solving equation (iii) and (iv), we get

$\therefore x = 1, \quad y = 2$

Centre is at $(1, 2)$

New constant, $c_1 = gx + fy + c$

$= (-12).1 + (-12).2 - 0$

$= -12 - 24$

$= -36$

Therefore, the equation is

$8x^2 + 4xy + 5y^2 - 36 = 0$

When the xy term is removed by rotation of axes. Let the reduced equation be

$a'x^2 + b'y^2 = 36$ … … … … … … … … … … … (v)

We know, $a' + b' = a + b = 8 + 5 = 13$

$\therefore a' + b' = 13$

$a'b' = ab - h^2 = 8.5 - (2)^2 = 36$

$(a' - b')^2 = (a' + b')^2 - 4a'b'$

$= 169 - 4.36$

$= 25$

$\therefore a' - b' = 5$

Now solving $a' + b' = 13$ and $a' - b' = 5$ we get

$a' = 9, b' = 4$

Putting the values a' and b' in (v), we get

$9x^2 + 4y^2 = 36$

$\Rightarrow \frac{x^2}{4} + \frac{y^2}{9} = 1$

$\Rightarrow \frac{x^2}{(2)^2} + \frac{y^2}{(3)^2} = 1$

The given equation represents an ellipse.

12. **Reduce the equation $16x^2 - 24xy + 9y^2 - 104x - 172y + 44 = 0$ to the standard form.**

Solution:

General equation of second degree is

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ … … … … … … … … … (i)

Given equation is

$16x^2 - 24xy + 9y^2 - 104x - 172y + 44 = 0$ … … … … … … … … … (ii)

Comparing equation (i) and (ii), we get

$a = 16, h = -12, b = 9, g = -52, f = -86, c = 44$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$\Delta = 16.9.(44) + 2(-86).(-52).(-12) - 16.(-86)^2 - 9.(-52)^2 - 44.(-12)^2$

$\Delta = 6336 - 107328 - 118336 - 24336 - 6336$

$\Delta = -250000$

$\Delta \neq 0$

Now

$ab - h^2 = 16.9 - (-12)^2$

$= 144 - 144$

$= 0$

$ab = h^2$

So this given equation can be represents a parabola.

The equation (ii) may be written as

$\therefore\ (4x - 3y)^2 = 104x + 172y - 44 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (iii)$

The lines $4x - 3y = 0$

and $104x + 172y - 44 = 0$

are not at right angles, hence let us introduce a constant λ in (iii)

$(4x - 3y + \lambda)^2 = (104x + 172y - 44) + 2\lambda(4x - 3y) + \lambda^2$

$(4x - 3y + \lambda)^2 = x(104 + 8\lambda) + y(172 - 6\lambda) + (\lambda^2 - 44) \ldots\ldots\ldots\ldots\ldots\ldots (iv)$

The lines $4x - 3y + \lambda = 0$

and $x(104 + 8\lambda) + y(172 - 6\lambda) + (\lambda^2 - 44) = 0$

are at right angles, if$(apply \quad a_1a_2 + b_1b_2 = 0)$

$4(104 + 8\lambda) + (-3)(172 - 6\lambda) = 0$

$416 + 32\lambda - 516 + 18\lambda = 0$

$50\lambda - 100 = 0$

$\therefore \lambda = 2$

Putting the value of λ in equation (iv), we get

$(4x - 3y + 2)^2 = x(104 + 8.2) + y(172 - 6.2) + \{(2)^2 - 44\}$

$\Rightarrow (4x - 3y + 2)^2 = x(104 + 16) + y(172 - 12) + 4 - 44$

$\Rightarrow (4x - 3y + 2)^2 = 120x + 160y - 40$

$\Rightarrow (4x - 3y + 2)^2 = 40(3x + 4y - 1)$

Which can be written as

$$\left(\frac{4x - 3y + 2}{\sqrt{4^2 + 3^2}}\right)^2 .25 = 40.5\left(\frac{3x + 4y - 1}{\sqrt{3^2 + 4^2}}\right)$$

$$\Rightarrow \left(\frac{4x - 3y + 2}{\sqrt{4^2 + 3^2}}\right)^2 = \frac{200}{25}\ \left(\frac{3x + 4y - 1}{\sqrt{3^2 + 4^2}}\right)$$

$$\Rightarrow \left(\frac{4x - 3y + 2}{\sqrt{4^2 + 3^2}}\right)^2 = 8\ \left(\frac{3x + 4y - 1}{\sqrt{3^2 + 4^2}}\right)$$

$$\Rightarrow \left(\frac{4x - 3y + 2}{\sqrt{4^2 + 3^2}}\right)^2 = 4.2\ \left(\frac{3x + 4y - 1}{\sqrt{3^2 + 4^2}}\right)$$

$Y^2 = 4AX$

Where

$$Y = \frac{4x-3y+2}{\sqrt{4^2+3^2}} \qquad X = \frac{3x+4y-1}{\sqrt{3^2+4^2}}, \quad A = 2$$

Which is the standard form of equation (ii).

13. **Reduce the equation $4x^2 - 4xy + y^2 - 8x - 6y + 5 = 0$ to the standard form.**

Solution:

General equation of second degree is

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$

Given equation is

$4x^2 - 4xy + y^2 - 8x - 6y + 5 = 0 \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (ii)$

Comparing equation (i) and (ii), we get

$a = 4, h = -2, b = 1, g = -4, f = -3, c = 5$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$\Delta = 4.1.5 + 2(-3).(-4).(-2) - 4.(-3)^2 - 1.(-4)^2 - 5.(-2)^2$

$\Delta = 20 - 48 - 36 - 4 - 20$

$\Delta = -88$

$\Delta \neq 0$

Now

$ab - h^2 = 4.1 - (-2)^2$

$= 4 - 4$

$= 0$

$ab = h^2$

So this given equation can be represents a parabola.

The equation (ii) may be written as

$\therefore\ (2x - y)^2 = 8x + 6y - 5 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (iii)$

The lines $2x - y = 0$

and $8x + 6y - 5 = 0$

are not at right angles, hence let us introduce a constant λ in (iii)

$(2x - y + \lambda)^2 = (8x + 6y - 5) + 2\lambda(2x - y) + \lambda^2$

$(2x - y + \lambda)^2 = x(8 + 4\lambda) + y(6 - 2\lambda) + (\lambda^2 - 5) \ldots\ldots\ldots\ldots\ldots\ldots . (iv)$

The lines $2x - y + \lambda = 0$

and $x(8 + 4\lambda) + y(6 - 2\lambda) + (\lambda^2 - 5) = 0$

are at right angles, if$(apply \quad a_1a_2 + b_1b_2 = 0)$

$2(8 + 4\lambda) + (-1)(6 - 2\lambda) = 0$

$16 + 8\lambda - 6 + 2\lambda = 0$

$10\lambda + 10 = 0$

$\therefore \lambda = -1$

Putting the value of λ in equation (iv), we get

$(2x - y - 1)^2 = x(8 - 4.1) + y(6 + 2.1) + \{(-1)^2 - 5\}$

$\Rightarrow (2x - y - 1)^2 = 4x + 8y - 4$

$\Rightarrow (2x - y - 1)^2 = 4(x + 2y - 1)$

Which can be written as

$$\left(\frac{2x - y - 1}{\sqrt{2^2 + 1^2}}\right)^2 . 5 = 4.\sqrt{5}\left(\frac{x + 2y - 1}{\sqrt{1^2 + 2^2}}\right)$$

$$\Rightarrow \left(\frac{2x - y - 1}{\sqrt{2^2 + 1^2}}\right)^2 = \frac{4.\sqrt{5}}{5} \left(\frac{x + 2y - 1}{\sqrt{1^2 + 2^2}}\right)$$

$$\Rightarrow \left(\frac{2x - y - 1}{\sqrt{2^2 + 1^2}}\right)^2 = 4.\frac{1}{\sqrt{5}} \left(\frac{x + 2y - 1}{\sqrt{1^2 + 2^2}}\right)$$

$Y^2 = 4AX$

Where

$Y = \frac{2x-y-1}{\sqrt{2^2+1^2}} \quad X = \frac{x+2y-1}{\sqrt{1^2+2^2}}, \quad A = \frac{1}{\sqrt{5}}$

Which is the standard form of equation (ii).

14 **Test the nature of the conic given by the equation $5x^2 - 24xy - 5y^2 + 4x - 58y - 59 = 0$.**

Solution:

General equation of second degree is

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$

Given equation is

$5x^2 - 24xy - 5y^2 + 4x - 58y - 59 = 0. \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (ii)$

Comparing equation (i) and (ii), we get

$a = 5, h = -12, b = -5, g = 2, f = -29, c = -59$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$\Rightarrow \Delta = 5.(-5).(-59) + 2.(-29).2.(-12) - 5.(-29)^2 + 5.(2)^2 + 59(-12)^2$

$\Rightarrow \Delta = 1475 + 1392 - 4205 + 20 + 8496$

$\Rightarrow \Delta = 7178 \neq 0$

$ab - h^2 = 5.(-5) - (-12)^2$

$= -25 - 144$

$= -169$

$ab - h^2 < 0$

$a + b = 5 + (-5)$

$= 5 - 5$

$= 0$

$\because \Delta \neq 0, a\,b - h^2 < 0, a + b = 0$

So the given equation represents a rectangular hyperbola.

15 Summary of the properties of ellipse with respect to its standard equation

The standard equation of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

(i) Co-ordinates of Centre; $(0,0)$

(ii) Co-ordinates of foci; $(\pm ae, 0)$

(iii) Co-ordinates of vertices; $(\pm a, 0)$

(iv) Equation of major axis; $y=0$

(v) Equation of minor axis; $x=0$

(vi) Equation of Directrix; $x=\pm\frac{a}{e}$

(vii) Length of major axis $=2a$

$(viii)$ Length of minor axis $=2b$

(ix) Eccentricity; $e^2=\frac{a^2-b^2}{a^2}$

(x) Length of Latus Rectum$=\frac{2b^2}{a}$

(xi) Equation of letera recta $x=\pm ae$

16 Summary of the properties of hyperbola with respect to its standard equation

The standard equation of an hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

(i) Co-ordinates of Centre; $(0,0)$

(ii) Co-ordinates of foci; $(\pm ae, 0)$

(iii) Equation of transverse axis; $y=0$

(iv) Equation of conjugate axis; $x=0$

(v) Equation of Directrix; $x=\pm\frac{a}{e}$

(vi) Length of the transverse axis $=2a$

(vii) Length of the conjugate axis $= 2b$

(vii) Eccentricity; $e^2 = \frac{a^2+b^2}{a^2}$

(ix) Length of Latus Rectum$= \frac{2b^2}{a}$

17 Summary of the properties of parabola with respect to its standard equation

The standard equation of an parabola $y^2 = 4ax$

(i) Co-ordinates of Centre; $(0,0)$

(ii) Co-ordinates of focus; $(a, 0)$

(iii) Equation of the directrix; $x + a = 0$

(iv) Equation of the axis; $y = 0$

(v) Length of the latus rectum $= 4a$

(vi) Equation of the tangent at the vertex $x = 0$

18 **Test the nature of the conic given by the equation $3x^2 - 8xy - 3y^2 + 10x - 13y + 8 = 0$.**

19 **Test the nature of the conic given by the equation $9x^2 - 24xy + 16y^2 - 18x - 101y + 19 = 0$.**

20 **Test the nature of the conic given by the equation $34x^2 + 24xy + 41y^2 + 48x + 14y - 108 = 0$.**

21 **Test the nature of the conic given by the equation $8x^2 + 4xy + 5y^2 - 16x - 14y + 13 = 0$.**

22 **Reduce the equation $4x^2 - 4xy + y^2 - 10y - 19 = 0$ to the standard form and identify it.**

23 **Reduce the equation $5x^2 - 6xy + 5y^2 + 22x - 26y + 29 = 0$ to the standard form and identify it.**

24 **Reduce the equation $8x^2 - 4xy + 5y^2 - 16x - 14y + 17 = 0$ to the standard form and identify it.**

25 **Reduce the equation $36x^2 + 24xy + 29y^2 - 72x + 126y + 81 = 0$ to the standard form and identify it.**